

A Micro Inertial Energy Harvesting Platform With Self-Supplied Power Management Circuit for Autonomous Wireless Sensor Nodes

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Abstract—A 0.25 cm³ autonomous energy harvesting micro-platform is realized to efficiently scavenge, rectify and store ambient vibration energy with batteryless cold start-up and zero sleep-mode power consumption. The fabricated compact system integrates a high-performance vacuum-packaged piezoelectric MEMS energy harvester with a power management IC and surface-mount components including an ultra-capacitor. The power management circuit incorporates a rectification stage with ~30 mV voltage drop, a bias-flip stage with a novel control system for increased harvesting efficiency, a trickle charger for permanent storage of harvested energy, and a low power supply-independent bias circuitry. The overall system weighs less than 0.6 grams, does not require a pre-charged battery, and has power consumption of 0.5 μ W in active-mode and <10 pW in sleep-mode operation. While excited with 1 g vibration, the platform is tested to charge an initially depleted 70 mF ultra-capacitor to 1.85 V in 50 minutes (at 155 Hz vibration), and a 20 mF ultra-capacitor to 1.35 V in 7.5 min (at 419 Hz). The end-to-end rectification efficiency from the harvester to the ultra-capacitor is measured as 58–86%. The system can harvest from a minimum vibration level of 0.1 g.

Index Terms—Energy harvesting, microelectromechanical devices, piezoelectric transducers, power management, wireless sensors.

I. INTRODUCTION

RAPID advances in solid-state devices over the past several decades have resulted in an increased functionality and performance in sensor systems, as well as a continual decrease in their power consumption. In conjunction with the lowered power requirements, recent advancements in energy conversion and power management technologies have enabled the realization of truly wireless and energy-independent sensor nodes, which harness energy from their environments instead of solely depending on an employed finite-lifetime battery or a costly power-line [1]. This energy autonomy is especially demanded in miniaturized wireless sensing applications, where

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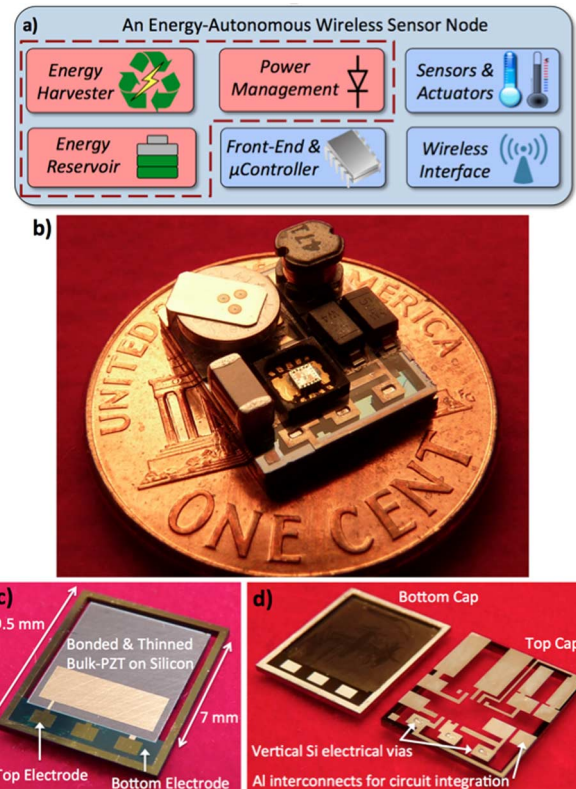


Fig. 1. (a) Fundamental elements of an energy-autonomous wireless sensor node. (b) The introduced self-supplied microsystem for inertial energy harvesting, management and storage. (c) A MEMS harvester fabricated with “bulk-PZT on silicon” technology. (d) Micro-fabricated silicon/glass dies used as bottom and top caps in the vacuum packaging of the MEMS harvester.

there is limited size and weight for long-term energy storage, and the battery maintenance is considered as impractical due to the location or packaging of the sensor node, or too costly due to the vast number of deployed nodes and frequent replacement needs. The critical components of a typical energy-autonomous wireless sensor are an energy harvester with sufficient power output, an efficient power management circuitry, and a compact and reliable energy reservoir (Fig. 1(a)).

Among alternative energy sources, there is an increasing interest in inertial harvesting, since energetic periodic vibrations can be found in the environment of some emerging applications, such as wireless industrial monitoring [2], [3], vehicle instrumentation [4] or structural health monitoring [5]. The maximum

theoretical power output that can be extracted from an ideal resonant vibration energy harvester is approximately [6]

$$Power_{MAX} = \frac{k^2 m Q^2 A^2}{8\pi f}. \quad (1)$$

According to this equation, an ideal 0.1 cm^3 sized inertial harvester, which has its resonance frequency (f) at 120 Hz, a bandwidth ($\sim f/Q$) of 10 Hz, 50% ratio of proof-mass (m) volume to device volume, and 10% electromechanical coupling (k^2), should theoretically be able to supply more than $100 \mu\text{W}$ when excited at resonance with 0.5 g acceleration amplitude (A). This level of power output could be sufficient for various low duty-cycle wireless sensor applications. In practice, to realize micro inertial harvesters with a similarly high power density, there are several optimization parameters. On the design side, one needs to maximize the effective mass and optimize the structural dimensions for higher mechanical coupling, and match the fundamental frequency of the ambient vibrations within the small device volume. On the fabrication side, it is necessary to incorporate a high energy-density material through a low-cost batch-mode process. Decreasing parasitic damping losses, and thus increasing the quality factor (Q), also improves the power density, although with a tradeoff in decreased bandwidth. Despite that the ambient vibrations in target industrial applications is mostly periodic, both the fundamental frequency of the vibration source and the resonance frequency of the harvester can experience small drifts over time due to fatigue, temperature or variable loading (e.g., change of speed or torque on a motor). Thus, in many cases, resonant devices with very narrow bandwidths (e.g., $<2 \text{ Hz}$) may not be very practical for long-term operations.

The most favorable technologies for transduction of vibration energy into electricity are based upon electrostatic, electromagnetic, and piezoelectric phenomena [7]. Electromagnetic harvesters have been favored due to their high reliability, simple assembly, and high efficiency at macro or meso scales [8]–[10]. A hand-assembled 0.15 cm^3 sized device has been shown in [11] to deliver $58 \mu\text{W}$ from 60 mg vibration at 52 Hz, as one of the harvesters with highest power densities in the literature, although with a bandwidth of $<0.5 \text{ Hz}$. However, further miniaturization or batch-mode micro fabrication of electromagnetic devices at a similar efficiency level is highly challenging due the required processes including fabrication of multiple layers of micro coil windings and integration of high-quality planar magnets on a wafer [12]. For instance, a wafer-level fabricated electromagnetic MEMS harvester with a similar size (0.13 cm^3) is demonstrated in [13] to harvest only $0.7 \mu\text{W}$ at 0.5 g vibration at 95 Hz. Due to their CMOS compatibility and standard microfabrication processing, electrostatic harvesters have been also widely studied [14], [15]. In order to improve the low efficiency of electrostatic devices compared to other transduction techniques, researchers have studied various methods, such as integrating an electret layer to increase charge density [16], voltage constrained charge extraction [17], optimizing structural dimensions [18], or using magnetic coupling to actuate electrostatic transducers [19]. Recently, a 1 cm^2 sized electret-based harvester with a corrugated electrode configuration is demonstrated to provide $160 \mu\text{W}$ at 2.9

g vibration at 728 Hz [20]. Compared to electrostatic and electromagnetic harvesters, piezoelectric transducers have advantages of high power density, easiness in scalability, and medium impedance and high voltage output that simplifies the design of a power management circuitry [21], [22]. A main constraint in the development of MEMS piezoelectric harvesters has been the limited material quality available from existing thin film deposition techniques, which provide lower energy conversion efficiency than the existing bulk materials [23]. In recent years, there has been a significant progress in both batch-mode microfabrication and power density of piezoelectric harvesters [24], [25]. In [26], an AlN based MEMS harvester is demonstrated to generate $85 \mu\text{W}$ at 1.75 g vibration and power a low duty-cycle wireless temperature sensor through a full-bridge rectifier in 1 cm^3 , although with a limited bandwidth of 2–3 Hz.

Challenges in the power management of harvested energy vary according to the utilized energy conversion technique due to different output impedances (a few Ω to 10s of $\text{M}\Omega$), voltage levels (mV to 10s of V), and operating frequencies (1 Hz to 10 kHz). A recent review of different rectification and power conditioning techniques is provided in [27]. Due to their capacitive source impedance, piezoelectric harvesters can suffer from reactive power loss and degradation in power transfer to an external resistive load. In order to increase harvesting efficiency from piezoelectric devices, some of the previous literature have focused on nonlinear electronic interfaces based on forming a resonant electrical network through brief connections to an off-chip inductor, such as synchronized switching (or resonant bias-flipping) [28], [29], synchronous charge extraction [30]–[32], and adaptive impedance matching [33] for continuous [34], [35] or periodic [36] maximum power point tracking. Basic principles and comparative analysis of nonlinear electronic interfaces for harvesters have been described in [9], [37]. Compared to standard rectification, depending on the electromechanical coupling and the mechanical damping of the harvester, these techniques can provide different benefits, such as improved energy extraction, boosting of input voltage, increased bandwidth, or decreased sensitivity to load impedance [37]. Although implemented in both board and IC level, these power management circuits have been only assembled and tested with commercial macro or meso scale harvesters, causing the overall system volume to be generally larger than 10 cm^3 .

The aim of this work is to realize an autonomous micro energy generation platform with competitive performance as a stand-alone unit that can be easily integrated with next generation wireless sensor nodes (Fig. 1(b)). Previous literature on micro inertial harvesters has focused on research efforts to improve the harvesting efficiency either only in the mechanical transducer or only in the power management circuitry. In this paper, a piezoelectric MEMS harvester with high power density is assembled in a compact system with an efficient integrated power management circuitry and off-chip energy reservoir units. In addition to high end-to-end mechanical-to-electrical conversion efficiency, the platform is aimed to start-up with initially discharged energy reservoirs for vibrational accelerations $> 0.1 \text{ g}$ ($1 \text{ g} = 9.8 \text{ m/s}^2$), consume low active-mode and zero sleep-mode power consumption from the harvested energy, and be able to harvest from low power inputs ($>3 \mu\text{W}$).

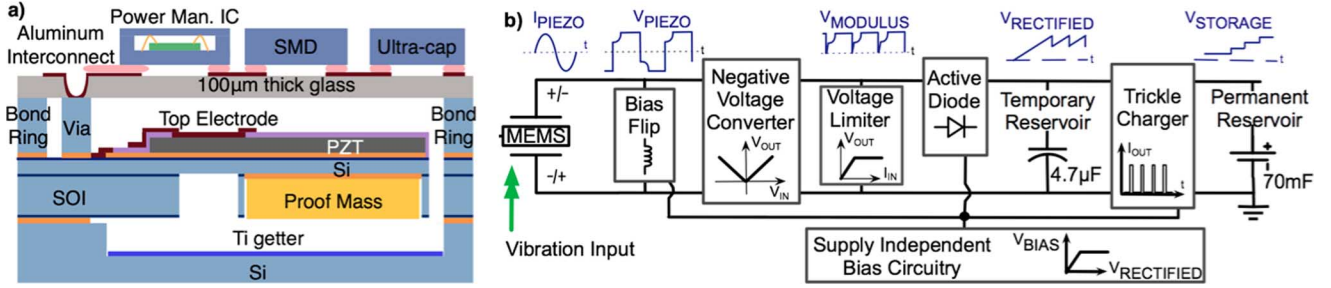


Fig. 2. (a) Cross-section of the inertial energy harvesting microsystem. (b) Overview of the power management circuit integrated with the MEMS harvester.

II. OVERVIEW OF THE ENERGY HARVESTING MICROSYSTEM

A. Vacuum-Packaged Piezoelectric MEMS Energy Harvester

The MEMS vibration harvester is realized with a recently developed process technology [38], [39], which offers advantages in microfabrication flexibility and larger electromechanical coupling over existing piezoelectric thin film deposition methods, such as AlN sputtering [40], PZT sputtering [41], or PZT sol-gel spin-coating [42]. This batch-mode fabrication technology involves aligned solder diffusion bonding and precision thinning of high-quality bulk-PZT ceramics on silicon [43]. The introduced process is performed at low temperature ($<200^{\circ}\text{C}$) and allows post-CMOS integration of a piezoelectric MEMS and its electrical circuitry on the same silicon die. In this work, hybrid integration is preferred due to the large difference in die sizes of the prototype harvester and the integrated circuit. The active device volume of the MEMS harvester is only 27 mm^3 (Fig. 1(c)), while still scavenging sufficient vibration energy for powering low power and low duty-cycle wireless sensor nodes with a presumed average consumption of $10\text{--}100\text{ }\mu\text{W}$ [44], [45]. The effective mass (m_{EFF}) and spring constant (k_M) of the harvester with tungsten proof mass are calculated as 330 milligrams and 377 N/m, respectively, and the mechanical damping coefficient (ζ_M) is measured as 0.0084 at open circuit. The electromechanical coupling is calculated as 0.085 according to $(k_{\text{SYS}}^2 = 1 - f_{\text{SC}}^2/f_{\text{OC}}^2)$ [6], where f_{OC} and f_{SC} are resonance frequencies measured from decaying damped oscillations due to an applied mechanical pulse to the transducer's base at open and short circuit configurations, respectively.

The fabricated MEMS harvester is hermetically packaged in a vacuum environment, and low-resistivity vertical silicon vias are used for electrical feedthroughs as shown in Fig. 2(a). The fabrication process of the silicon-glass top cap is adapted from [46], where a similar technique has been used for packaging of inertial sensors. To allow free displacement of the harvester's proof mass under external vibration, both the silicon bottom cap and the silicon-glass top cap incorporate a recess of $>300\text{ }\mu\text{m}$. In addition, the bottom cap has a $2\text{ }\mu\text{m}$ thick sputtered titanium layer in order to facilitate pumping out the gaseous molecules trapped in the cavity with a post-packaging heat treatment [47]. On top of the glass substrate of the device package, there is a $1\text{ }\mu\text{m}$ thick sputtered and patterned aluminum layer for signal routing, on which the power management IC and the surface-mount device (SMD) components are assembled through soldering or Ag-epoxy (Fig. 1(d)). The total volume of the pack-

aged generator with its circuitry and all SMD components is less than 0.25 cm^3 , and weighs 0.6 grams while incorporating a tungsten proof mass. Prior to integration with a wireless sensor, this autonomous energy generation platform is to be covered with a thermally conductive epoxy for mechanical protection, and include just two power leads from the ultra-capacitor to the outside world.

B. Integrated Circuit and Energy Storage Components

In order to provide fully autonomous operation, the power management circuitry is designed to be self-supplied by the harvested vibration energy, and have no dependency on a previously-charged energy reservoir. The circuitry consists of three main sub-circuits (Fig. 2(b)). First, a bias-flip stage is used to increase the harvesting efficiency from the piezoelectric device. Then, a cross-coupled CMOS rectifier and an active diode rectify the harvester output with minimum voltage drop. Finally, a trickle battery charger transfers the collected energy in the temporary reservoir to a permanent reservoir through intermittent current pulses, and charges the permanent reservoir up to a pre-defined voltage level. Although a DC-DC regulator can also be connected to this permanent energy reservoir, it is not included in this prototype since the fluctuation in the terminal voltage of the reservoir due to an external load is expected to be tolerable. For instance, a peak energy consumption of 6 mJ by an intermittent radio activity [48] will cause only a small voltage decrease from 1.85 V to 1.80 V on a 70 mF reservoir. Under sufficient excitation, the vibration harvester is expected to fully recharge the permanent reservoir until the next energy-consumption cycle by the load.

The designed power management IC is realized utilizing the TSMC high-voltage $0.18\text{ }\mu\text{m}$ technology (Fig. 3(a)). The active device area occupies 0.25 mm^2 , where the greatest portion is spent for on-chip resistors and diodes. The fabricated CMOS chip is wire-bonded to a quad flat package (a surface mount plastic package with lead pads located on the bottom surface of the package), which is epoxied/soldered to the top cap of the MEMS harvester. The off-chip components include a $4.7\text{ }\mu\text{F}$ ceramic chip capacitor as the temporary energy storage, an ultra-capacitor (70 mF or 20 mF) as the permanent energy storage, and two Schottky diodes (BAT54WS-TP) and a $470\text{ }\mu\text{H}$ inductor for the bias-flip stage (Fig. 3(b)). The 25 mm^3 sized coin SMD ultra-capacitor utilizes electrochemical double layer capacitor (EDLC) construction, and has an operating voltage from 0 V to 3.3 V [49]. Despite its larger self leakage, an ultra-capacitor suits the storage of the harvested energy better than a

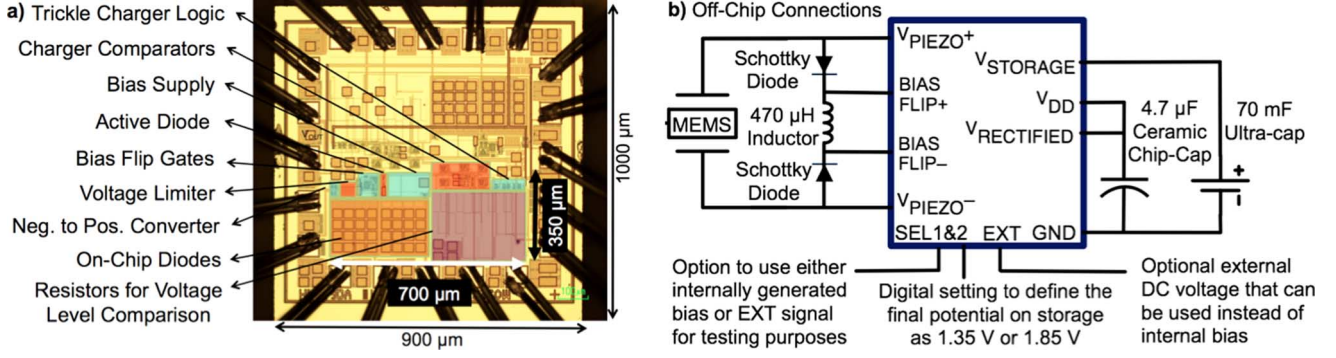


Fig. 3. (a) Micrograph of the power management IC die, and reserved areas for sub-circuits. (b) Off-chip connections of the IC to the surface mount components.

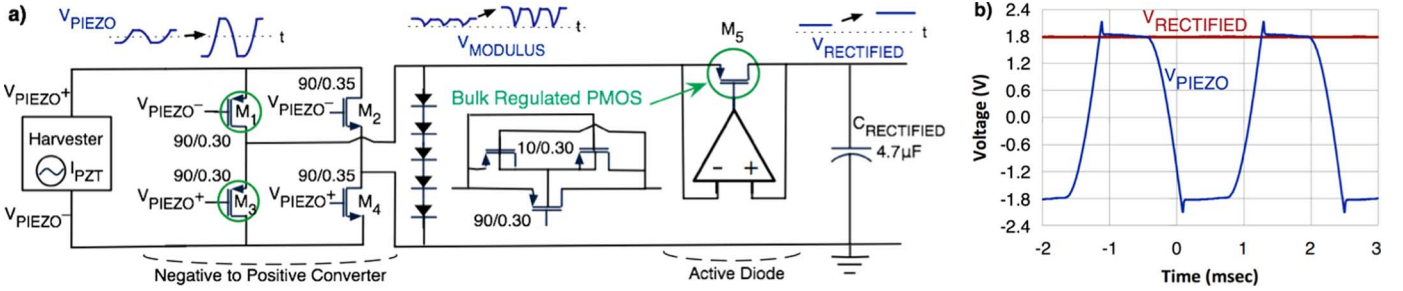


Fig. 4. (a) Two-stage rectification to minimize voltage drop on the signal path from the harvester to the temporary energy reservoir. (b) Measured voltage drop between the harvester output and the rectified signal in the temporary reservoir.

rechargeable battery, since it provides a much longer life-time with little degradation over a few million charging cycles compared to a few hundred cycles available from most rechargeable batteries. Also, a high power density becomes more essential than a high energy density in self-powered low-duty-cycle sensor nodes, where the sudden discharge current limitation of $\sim 5 \mu\text{A}$ in a 25 mm^3 rechargeable battery may not be sufficient to meet the intermittent high current draw of a wireless system [50].

III. POWER MANAGEMENT CIRCUIT DESIGN

A. Active Rectification With Low Voltage Drop

The rectification of the sinusoidal current output from the piezoelectric MEMS harvester is achieved in two stages (Fig. 4(a)), with a similar topology used in [51] for electromagnetic harvesters. First, four cross-coupled CMOS switches output the modulus (V_{MODULUS}) of harvester potential (V_{PIEZO}) by converting the negative half cycles into positive ones [52]. According to the input polarity, either M_1 and M_4 , or M_2 and M_3 are pairwise conducting, so that V_{MODULUS} is always connected to the higher voltage potential of the MEMS harvester. Since the gates are driven by a high voltage swing, the voltage drop across this stage depends only on the on-resistance of the MOS switches instead of the threshold voltage (V_T) as in standard diode-connected MOS transistors. However, this stage cannot be used alone for rectification, since it does not block the reverse current. Thus, an active diode is utilized at the output to rectify the voltage and store the harvested charge on a temporary energy reservoir ($V_{\text{RECTIFIED}}$), which also serves as the power supply of the active circuitry (V_{DD}).

The active diode turns on with a near-zero voltage drop whenever V_{MODULUS} is higher than $V_{\text{RECTIFIED}}$, and blocks the reverse current flow when V_{MODULUS} drops below $V_{\text{RECTIFIED}}$. The op-amp in the active diode enables supply independent positive rail operation, and senses the polarity change across the difference between $V_{\text{RECTIFIED}}$ and V_{MODULUS} despite a high common-mode voltage (Fig. 5(a)) [53]. When the non-inverting input (V_+) is at a higher potential than the inverting input (V_-), comparator output saturates at V_{MODULUS} , opening the gate M_5 . When the non-inverting input (V_+) drops below the inverting input (V_-), the output saturates at ground, closing the gate M_5 . In order to minimize back leakage, the PMOS gate M_5 is bulk-regulated such that its n-well is always connected to the highest potential available from its source and drain nodes [54]. An offset voltage is added to the op-amp design ($V_{\text{OFFSET-1}}$, $\sim 15 \text{ mV}$) against process tolerances by mismatching the aspect ratios of PMOS transistors connected to the negative and positive terminals. This offset value prevents a possible backflow of charge from the temporary reservoir during the bias-flip, since the comparator used in the bias-flip stage has a faster switching time than the comparator used in the active diode.

Compared to ideal-diode rectification, the active diode requires some power consumption for timing control of M_5 . However, to enable low power operation, the active diode can be designed to have a small bandwidth, since the MEMS harvester operates at less than 500 Hz. When biased at 530 mV, the comparator consumes only 34 nA, while providing a sufficiently fast signal switching of 48 μs (Fig. 5(b)).

Overall, the two-stage active rectification scheme minimizes the voltage drop on the path between V_{PIEZO} and $V_{\text{RECTIFIED}}$

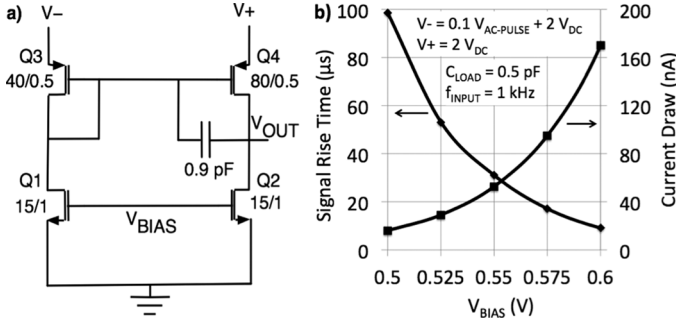


Fig. 5. (a) Comparator design used in the active diode. (b) Dependence of current consumption and signal rise time of the comparator on bias voltage.

to only ~ 30 mV (Fig. 4(b)). The minimum input voltage from the harvester required to drive the initial cross-coupled CMOS gates for low loss rectification depends on the threshold voltage of the transistors, which is ~ 0.5 V for the TSMC $0.18 \mu m$ technology. This threshold is not a significant limitation for the presented system, since at an input vibration of 0.1 g, the MEMS harvester delivers more than 0.9 V or 1.6 V on a resistive load when it incorporates a silicon proof mass or tungsten proof mass, respectively. Finally, a voltage limiter is also implemented between the negative voltage converter and the active diode, by a series of on-chip diodes connected between $V_{MODULUS}$ and the ground line. This precaution allows for protection of the circuit against possible high power peaks from the harvester in case of a mechanical shock, and limits $V_{MODULUS}$ to < 3.3 V.

B. Supply-Independent Bias Generator

A supply-independent bias circuitry is adapted from [32] and [55], and provides a steady voltage of ~ 530 mV under varying supply voltages from the temporary reservoir for $V_{RECTIFIED} > 0.6$ V (Fig. 6). This output voltage is used to bias all the comparators in the system, and provide a constant current flow across them despite the oscillating $V_{RECTIFIED}$. This limits the overall power consumption, since the power becomes linearly dependent on the supply voltage. The generated V_{BIAS} is also used as the reference voltage (V_{REF}) in the trickle charger to define the voltage regulation levels (V_1 – V_4) during charging of the permanent energy reservoir. Since the generated bias is prone to be affected by threshold variations due to process mismatches, this can cause slight variations in the final potential on the ultra-capacitor over different chips. In order to provide a more stable supply voltage in a particular application, a low-dropout regulator based on bandgap reference can be added between the output of ultra-capacitor and the load circuitry.

C. Low-Power Bias-Flip Stage With Relaxed Timing Control

The piezoelectric MEMS harvester can be electrically modeled as a Norton equivalent circuit consisting of a variable current source (I_{PZT}) dependent on the mechanical vibration, the PZT self-capacitance (C_{PZT}) and the dielectric loss (R_{PZT}) (Fig. 7(a)). This simplified model is used here for basic description of the bias-flip operation. A more accurate equivalent circuit model of a piezoelectric harvester can be found in [56],

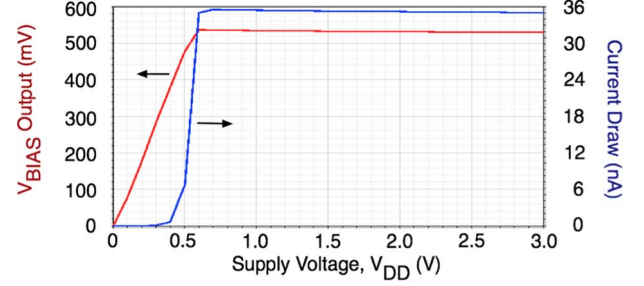


Fig. 6. The simulated constant current consumption and voltage output of the bias generator for supply voltages 0.6 V to 3 V.

which includes the electromechanical feedback from the circuit to the harvester affecting the electrical damping and resonance frequency based on the electrical load. A conventional rectifier can harvest only some portion of the piezoelectric current output, since the internal capacitor of the harvester reduces the available power transferred to the rest of the circuit. This is because at each half-cycle, when the harvester's current output changes polarity, there is energy wasted to discharge C_{PZT} , and charge it back in reverse polarity (Fig. 7(a)). This loss can be partially decreased by shorting C_{PZT} temporarily whenever the piezoelectric current crosses zero in either direction (*shunt-pass*), allowing the neutralization of the accumulated electric charge for a brief period [57]. With this method, the time period required to reverse the voltage polarity is decreased, $|V_{PIEZO}|$ can reach to the level of $V_{RECTIFIED}$ faster, and thus the rectifier can harvest a greater portion of the generated piezoelectric charge.

A more efficient method to reduce the charge loss in a capacitive current source is *resonant bias flipping* (also named as synchronized switch damping, or parallel SSHI). Here, an external inductor, connected in parallel to the harvester for intermittent periods, forms a resonant LC tank temporarily and redirects the energy in C_{PZT} to charge itself in reverse polarity. This technique has been first proposed in [58] to enhance semi-passive damping of a piezoelectric actuator for vibration isolation. Later, it is adapted for energy harvesting purposes in [59] and [60]. An analytical study of the increased harvesting efficiency by bias flipping is provided in [61] and [37]. The effect of frequency variation on the efficiency of this nonlinear circuit interface has also been investigated previously in [62]. For weakly and moderately coupled harvesters, the SSHI interface provides a benefit of increased bandwidth, while for strongly coupled harvesters ($k^2/\zeta_M > 10$) it does not improve or degrade the bandwidth of the harvester.

A more detailed explanation of bias flipping is as follows. At the end of a half-cycle when I_{PZT} changes polarity and V_{PIEZO} starts its decay from its maximum value, an external inductor is connected across the harvester (Fig. 7(b)). The stored charge on C_{PZT} is rapidly depleted through the inductor until V_{PIEZO} reaches zero. However, the charge flow builds up a magnetic field across the inductor, which opposes sudden changes in current by Lenz's Law. Therefore, the current continues to flow through the LC tank in the same direction until the magnetic field is completely dissipated. At this point, the original charge on C_{PZT} is restored, but in opposite polarity and with some loss

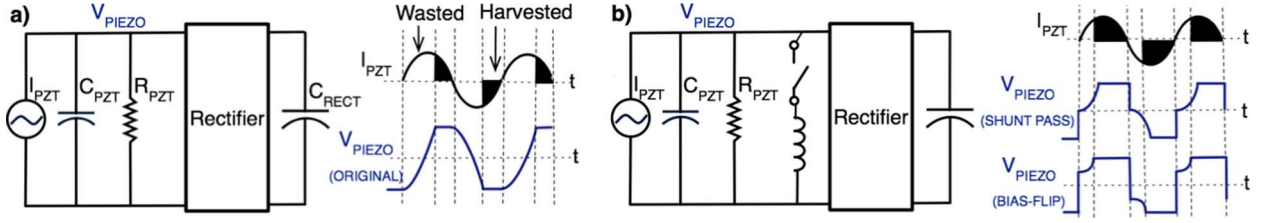


Fig. 7. Graphical comparison of harvested power (a) with normal rectification, and (b) with rectification including shunt pass or bias flip operations.

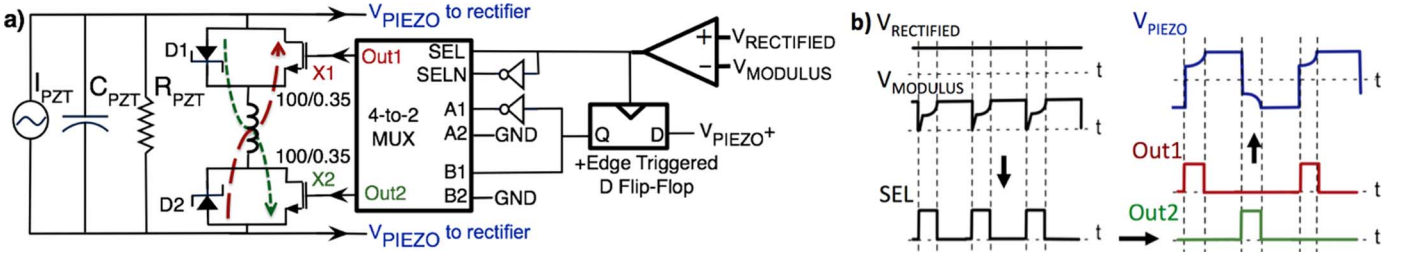


Fig. 8. (a) The control circuitry used to drive the gates in the bias-flip stage. (b) Auto-timing of bias-flip gates independent of the chosen inductor or harvester.

due to device non-idealities. As V_{PIEZO} is now in the same polarity with I_{PZT} , the inductor has to be abruptly disconnected from the harvester at this time, otherwise the harmonic cycle would continue with the LC tank current flowing in the opposite direction. This back flow of charge can be prevented by timing the bias-flip switch for only one half-period of LC tank oscillation as previously demonstrated in [29]. However, this method of precision timing requires a fast and power-consuming control circuitry and manual tuning for different C_{PZT} or inductor values.

Here, this work introduces a low-power controller for a bias-flip stage (Fig. 8(a)). In order to time the driving of the switches that control the inductor path, the circuitry detects the polarity change in the piezoelectric current by comparing $V_{MODULUS}$ and $V_{RECTIFIED}$ (Fig. 8(b)). Whenever $V_{MODULUS}$ drops below $V_{RECTIFIED}$, either the X_1 or the X_2 gate is turned on, depending on the polarity of the current half-cycle, determined by an edge-triggered flip-flop. The switch remains on during the whole C_{PZT} recharging period and until $V_{MODULUS} > V_{RECTIFIED} - V_{OFFSET}$, while a Schottky diode (D_1 or D_2) prevents the backflow of charge after one half-period of LC tank oscillation. The reverse current across the diodes is ~ 50 nA, and results in only $< 0.04 \mu W$ leakage loss. Although the Schottky diodes involve some additional power loss due to their forward voltage drops (~ 0.15 V) during bias-flipping, they eliminate the requirement of fast-switching and power-consuming active diodes on the LC-tank signal path. Compared to previous works on resonant bias-flipping, the introduced design does not require a manually fine-tuned timing control depending on different inductor values as in [29] or a tuned differentiator designed for a specific harvester for pre-determined vibration amplitude and frequency as in [63]. The measured and overlapped voltage waveforms at the output of the fabricated harvester are presented in Fig. 9, when standard rectification, shunt pass or bias flipping is employed for the same vibration input. The capacitance of the

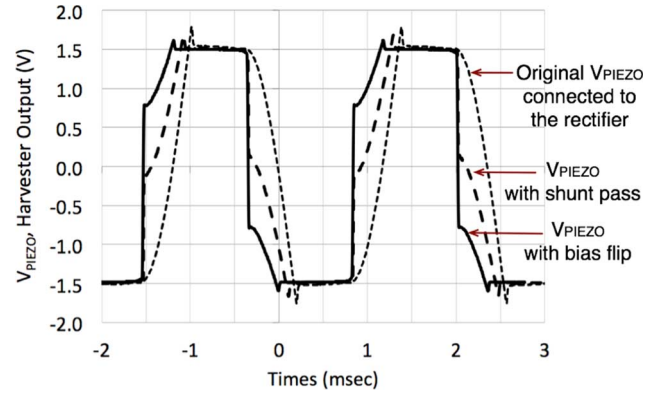


Fig. 9. The measured and overlapped voltages from the piezoelectric harvester with standard rectification, with shunt pass and with bias flip.

harvester is measured as ~ 8.5 nF, and a $470 \mu H$ inductor is used in the bias-flip stage.

D. Trickle Charging of Permanent Energy Reservoir

A trickle charger is designed in order to intermittently transfer the collected charge in the temporary reservoir ($C_{RECTIFIED}$) to a permanent reservoir ($C_{STORAGE}$), which could be either an ultra-capacitor or a re-chargeable battery (Fig. 10(a)). The operation principle is illustrated in Fig. 10(b), where the system starts off with no initial charge stored in either reservoirs. At this point, the bulk-regulated PMOS gate P_1 isolates the path between $C_{RECTIFIED}$ and $C_{STORAGE}$. When there is sufficient vibrational excitation, the MEMS harvester is able to supply enough current to charge $C_{RECTIFIED}$ passively up to 0.8 V, the minimum voltage required for active rectification to turn on. This initial passive charging is enabled by the body-drain diode of M_5 in the active diode. After this start-up period, $C_{RECTIFIED}$ continues to be charged up, but now more efficiently with the bias generator, the active diode and the bias flip stage fully operational. As $V_{RECTIFIED}$ reaches to a

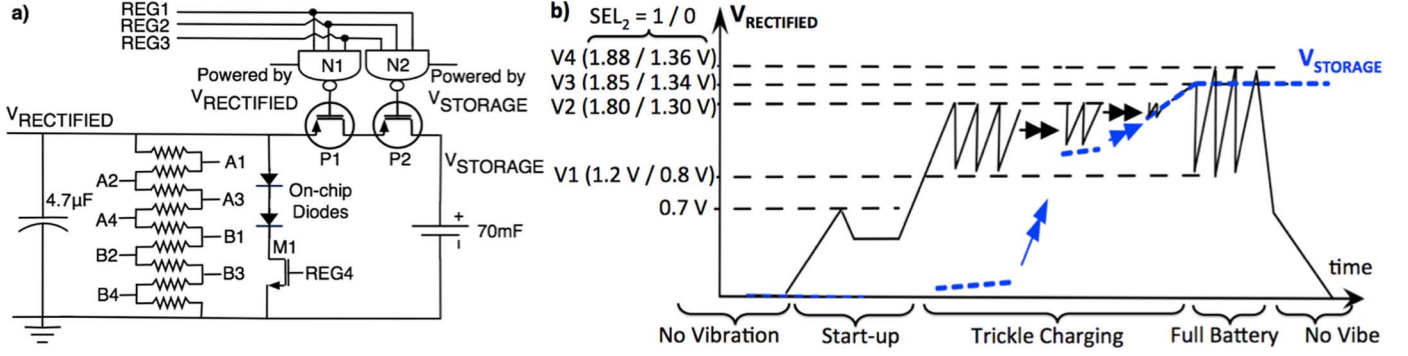


Fig. 10. (a) The trickle charger for transferring the scavenged energy into the permanent reservoir. (b) Graphical representation of the trickle charging of a permanent reservoir with zero initial charge, while the active circuitry is self-supplied from the harvested energy in the temporary reservoir.

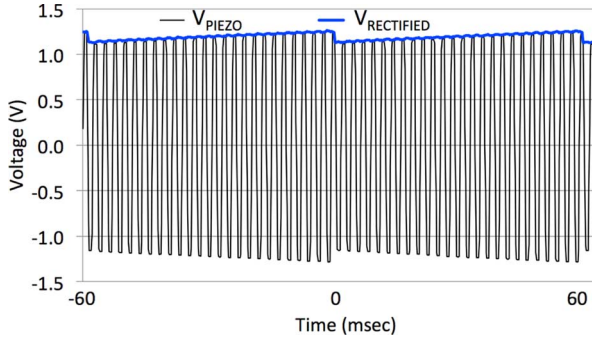


Fig. 11. Measured voltage signals at the output of the piezoelectric harvester (V_{PIEZO}) and on the temporary energy reservoir ($V_{RECTIFIED}$) during operation.

certain potential (V_2), both of the power gates P_1 and P_2 turn on temporarily, and part of the scavenged energy is transferred from $C_{RECTIFIED}$ to the battery/ultra-capacitor through a current pulse until $V_{RECTIFIED}$ drops down to the level of either V_1 or $V_{STORAGE}$. Then, P_1 is turned off, and $C_{RECTIFIED}$ is charged up again back to the potential of V_2 . This intermittent battery-charging scheme continues till $V_{STORAGE}$ reaches to the V_3 potential (Fig. 11). At this point, the power gates turn off not to overcharge the battery/ultra-capacitor above its rated voltage. In order to avoid the control system getting locked at this stable operation point, $V_{RECTIFIED}$ is put into scanning between the V_1 and V_4 potentials by draining the extra harvested charge to the ground through M_1 . This allows the system to continuously check whether $V_{RECTIFIED}$ is dropped below V_2 and if the battery needs to be recharged again. When the input vibration ends, $V_{RECTIFIED}$, which is also utilized as the power supply (V_{DD}), decays down to zero, and the active circuitry turns off. The back flow of battery charge to the temporary reservoir is prevented by P_2 , which is controlled via a NAND gate (N_2) powered by the battery. During this sleep mode operation state, the power consumption of the system only comes from N_2 and the near-zero leakage of $C_{STORAGE}$ back to the $C_{RECTIFIED}$, which results in an overall very low stand-by current draw of <5 pA.

To determine the voltage regulation levels (V_1 – V_4), pre-determined resistive fractions (C_1 – C_4) of $V_{RECTIFIED}$ are compared with a stable reference signal V_{REF} in the control circuitry of trickle charger (Fig. 12). The resistors are integrated on the

chip, have a total value of ~ 25 M Ω in series, and thus occupy a significant part of the layout. The combination of output signals REG_{1-3} provide the on/off control of the charging power gates between the temporary and permanent storage. The intermittent charging of $V_{STORAGE}$ is enabled by the REG_1 control signal, while the REG_2 signal is used to make sure that the power gates are closed only if $V_{RECTIFIED}$ is higher than $V_{STORAGE}$ so that the backflow of charge is avoided. The REG_3 signal prevents overcharging of $C_{STORAGE}$ over its rated potential, and the REG_4 signal regulates the oscillation on $C_{RECTIFIED}$ between V_1 and V_4 potentials once the battery is fully charged (scanning mode).

In order to allow more flexibility in testing, some additional control signals are also utilized in the circuit. In standard autonomous operation, the reference signal (V_{REF}) used in the comparators is obtained from the internally generated V_{BIAS} . However, an external signal (V_{EXT}) can also be used for V_{REF} by setting SEL_1 , so that the final charging level on $V_{STORAGE}$ can be adjusted up to 3 V, the highest gate potential allowed in the utilized CMOS process (Fig. 12). Alternatively, the set of resistors that define the C_1 – C_4 comparison signals from the fractions of $V_{STORAGE}$ can be altered by setting SEL_2 , and the final charging level can be chosen as either 1.35 V or 1.85 V.

E. Low-Power Comparator for Bias-Flip and Trickle Charger

Both the bias flip stage and the trickle charger require a low-power comparator design that can still provide sufficiently fast switching. To meet the requirements, a three-stage comparator is leveraged (Fig. 13(a)). The first stage in the comparator is a common-source differential amplifier with a single-ended output. Here, T_1 provides a constant current source with a small voltage drop across it, and loaded by two branches that are connected in parallel for signal-amplification. The transistors T_4 and T_5 provide active current mirroring between the left and right branches, and allows to obtain a single-ended output while canceling out the common mode signal from the inputs. Even though a perfect cancellation of common mode is not possible with this design as in the case of a fully differential amplifier, it still provides a high common-mode rejection ratio to obtain an effective comparison of input signals with possibly large off-set voltages. The current mirror across T_4 and T_5 inverts the small signal current flowing through T_2 and passes it across T_3 . At the output between T_3

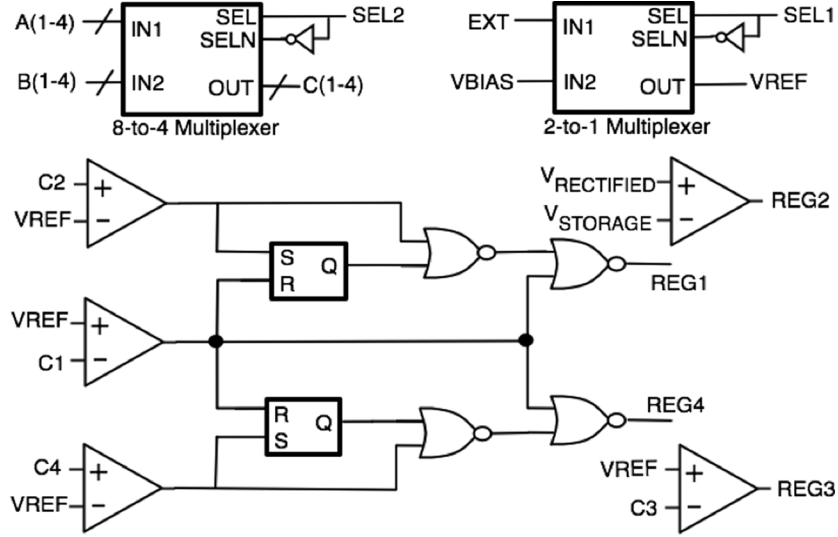


Fig. 12. Control circuitry of the trickle charger to time the on/off switching of the charging gates between the temporary and permanent energy reservoirs.

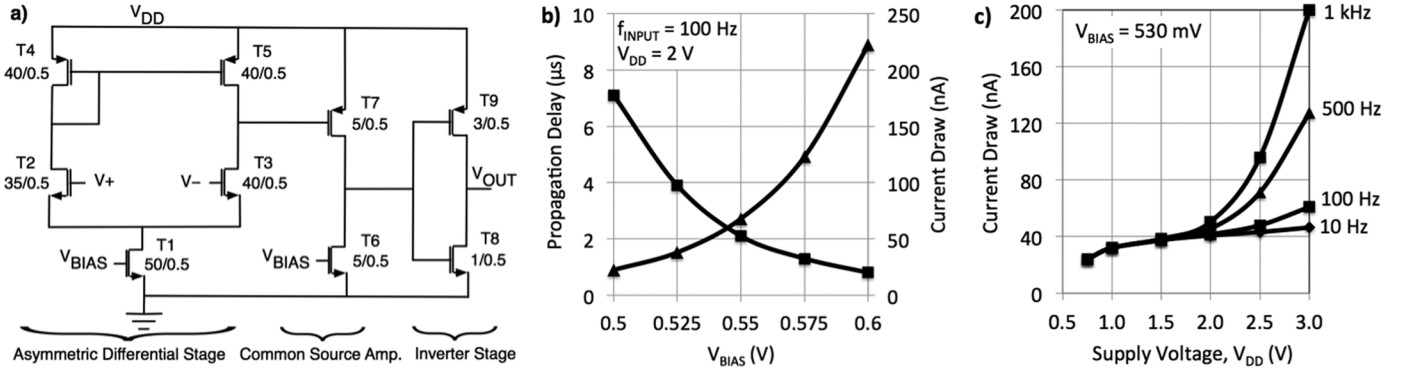


Fig. 13. (a) Three-stage comparator design for the bias-flip stage and trickle charger. (b) Simulated trade-off between signal propagation delay and current consumption at different V_{BIAS} . (c) Simulated power consumption of the comparator with increasing V_{DD} ($= V_{RECTIFIED}$) and with different input frequencies.

and T_5 , the signal currents from the inputs ($V+$ and $V-$) are subtracted.

The small signal gain from the differential input to the single-ended output can be calculated approximately as $0.5 \times (g_{m2} + g_{m3}) \times (r_{o3} // r_{o5})$, where g_m is the transconductance and r_o is the output resistance of a transistor. If T_2 and T_3 have equal dimensions, their transconductance values would be equal. However, a small asymmetry is added between the transistors in order to provide an offset of ~ 15 mV ($V_{OFFSET-2}$) in the comparator's signal transition. While driving the bias-flip stage, this offset causes the comparator to wait until $V_{MODULUS}$ drops below $(V_{RECTIFIED} - V_{OFFSET-2})$ before switching its output. This precaution makes sure that enough time has passed after the active diode's complete turn off, and it is safe for the comparator to turn on the bias-flip gates. Otherwise, if the active diode is still on when a bias-flip gate is turned on, some of the rectified charge from the temporary storage will leak back into the bias-flip stage, and decrease the overall circuit efficiency.

The output of the differential amplifier is connected to a single-input single-output common source amplifier in order to provide high voltage swing at the output. Here, T_6 operates at saturation region acting as a current source load, and the

gain from this stage is approximately $g_{m7} \times (r_{o6} // r_{o7})$. Since the described amplifier is to be used for signal comparison, an important parameter is the slope and speed of rail-to-rail output transition. In order to improve the limited slew rate from the two-stage amplifier, an inverter is added at the end, which provides a binary-state digital output with fast transition. The transistor sizes and V_{BIAS} of the comparator are adjusted to minimize the signal propagation delay, so that the bias-flip stage can operate much faster than the input vibration frequency for maximum harvesting efficiency (Fig. 13(b)). The current consumption of the comparator is simulated to be less than 40 nA for a supply voltage < 2 V and a switching frequency < 1 kHz (Fig. 13(c)).

IV. SYSTEM-LEVEL MEASUREMENTS

The overall autonomous energy-scavenging microsystem, including the MEMS harvester and the power management circuitry, is tested on a shaker table, and excited at the mechanical resonance frequency of the harvester at different acceleration levels (Fig. 14(a)). The system is tested for charging different values of capacitors and ultra-capacitors, since the choice of permanent energy reservoir can affect both the overall efficiency of charge transfer from the harvester into the storage unit and the

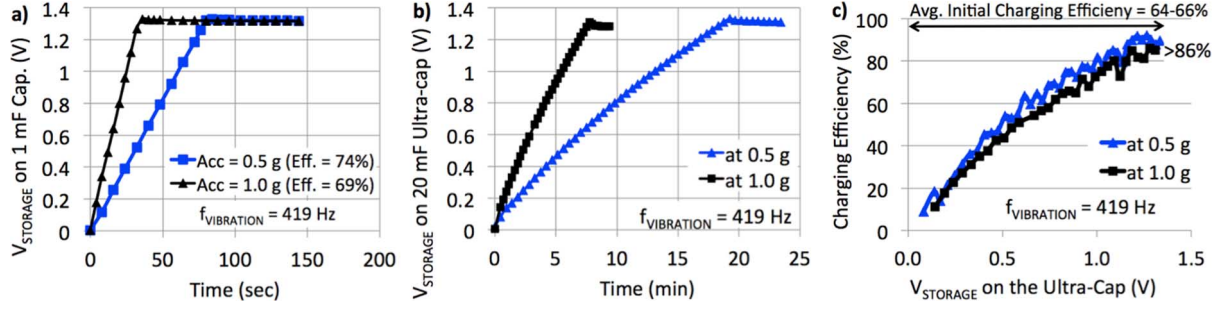


Fig. 14. (a) Charging of a 1 mF tantalum chip capacitor under different vibration levels. (b) Charging of a 20 mF ultra-capacitor with a MEMS harvester incorporating a silicon proof mass excited at 419 Hz vibration. (c) Effective charging efficiency of 20 mF ultra-capacitor, calculated from data in Fig. 14(b).

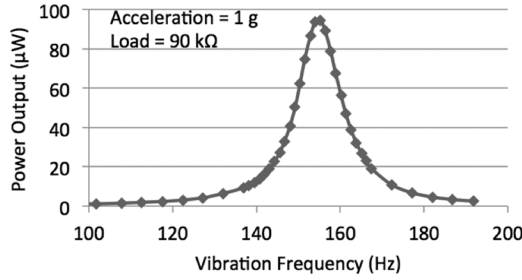


Fig. 15. Raw power output from the MEMS harvester with tungsten proof mass when excited with 9.8 m/s² acceleration.

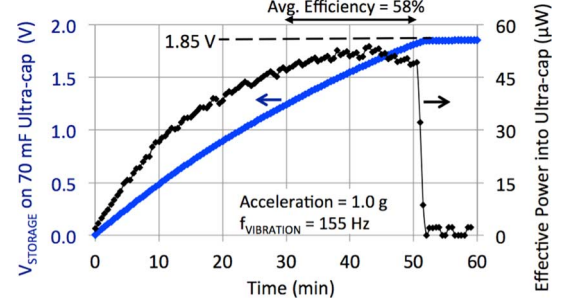


Fig. 16. Autonomous charging of an initially depleted 70 mF ultra-capacitor by MEMS harvester with tungsten proof mass excited at 1 g at 155 Hz.

energy leakage inside the storage unit in long term. The charging efficiency is calculated by taking the ratio of effective power into the ultra-capacitor over the raw power output from the harvester.

$$\text{Efficiency} = \frac{\text{Effective Power}}{\text{Raw Power on } R_{\text{OPTIMUM}}} \quad (2)$$

$$\text{Effective Power} = \frac{C_{\text{STORAGE}}(V_{\text{FINAL}}^2 - V_0^2)}{2(t_{\text{FINAL}} - t_0)} \quad (3)$$

When a MEMS harvester with silicon proof mass is connected to its optimum resistive load without the power management circuitry, it is measured to supply 24.1 μW (6.1 V_{PP}) and 63.9 μW (10 V_{PP}) at 0.5 g and 1.0 g vibration levels, respectively. When this harvester is connected with its power management IC, it can charge a 20 mF ultra-capacitor up to 1.35 V in 19 min under 0.5 g vibration, and in 7.5 min under 1.0 g vibration input (Fig. 14(b)). The first-time startup charging efficiency is measured as ~64% on the average during the whole charging period from 0 V to 1.35 V. The efficiency is maximized to 86% towards the end of charging when the potential on C_{STORAGE} is closer to $V_{\text{RECTIFIED}}$ (Fig. 14(b)). It is observed that there is a decrease in efficiency at higher vibration accelerations, and this is attributed to the fact that the optimum harvester voltage for maximum power output is increasing while the actual V_{PIEZO} is limited by the pre-defined final V_{STORAGE} value. Compared to charging a 1 mF tantalum capacitor, the harvesting efficiency was observed to be higher for the 20 mF ultra-capacitor case, presumably due to the lower internal series resistance of this component against pulse-charging. When the circuitry operates

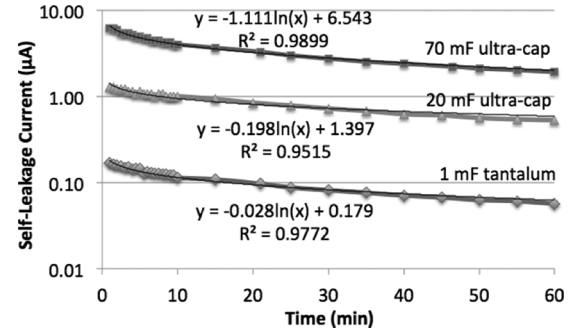


Fig. 17. Self-leakage currents of different types of capacitors that are used as alternative permanent energy storage units.

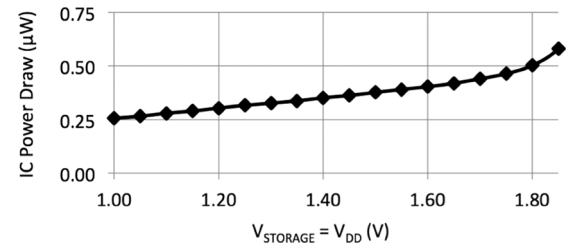


Fig. 18. Power consumption of the power management IC from the temporary reservoir during active operation.

with the resonant bias-flipping, the maximum efficiency is measured to be only 2.6% and 4.9% higher compared to rectification with shunt-pass and rectification without any bias-flip or shunt-pass, respectively. This limited improvement in the efficiency is caused by the moderate to high level of electromechanical coupling in the piezoelectric harvester, and the increased

TABLE I
SUMMARY OF THE SELECTED STATE-OF-THE-ART PIEZOELECTRIC AND ELECTROMAGNETIC ENERGY HARVESTING PLATFORMS.

Ref.	Electromechanical Harvester					Power Management Circuit					System
	Device Technology	Frequency, Bandwidth	Raw Power	Power Density (Unpackaged)	Unpackaged Volume	Conversion Architecture	Process, Chip-area	Power Consumption	Charging Efficiency	Start-up requirements	
This Work	Piezoelectric (MEMS)	155 Hz (13 Hz) 419 Hz (26 Hz)	95 μ W (1.0 g) 64 μ W (1.0 g)	3.5 mW/cm ³ /g ² 2.4 mW/cm ³ /g ²	27 mm ³	NPVC, Active diode, Bias-flip, Trickle charger	0.18 μ m CMOS, 0.3 mm ²	0.5 μ W (active) 10 pW (sleep)	58-86%	Self-powered P _{IN} > 2.7 μ W	0.25 cm ³ A _{IN} > 0.1 g V _O =1.8 V
[26]	Piezoelectric (MEMS)	325 Hz (3 Hz)	85 μ W (1.75 g)	0.97 mW/cm ³ /g ²	29 mm ³	Schottky rectifier, (+ MCU, Radio)	Discrete components	— (10 μ W entire system)	60%	Self-powered —	1 cm ³ A _{IN} > 0.6 g V _O =1.8 V
[29]	Piezoelectric	225 Hz (—)	— (3.35 g)	—	130 mm ³	Diode bridge, Bias-flip, DC-DC buck	0.35 μ m CMOS, 4.3 mm ²	2 μ W	70-85%	Battery (1.8 V) P _{IN} > 10 μ W	— V _O =1.8 V
[32]	Piezoelectric	100 Hz (—)	40 μ W (0.016 g)	—	229 mm ³ + tip mass	Active diodes AC-DC step-up with SECE	2 μ m BiCMOS, 0.9 mm ²	0.9 μ W	75%	Battery (2.5 V) P _{IN} > 4 μ W	— A _{IN} > 5 mg V _O =4.5 V
[64]	Piezoelectric	174 Hz (—)	557 μ W (0.8 g)	—	138 mm ³ + tip mass	NPVC, Active diode, Series SECE	0.35 μ m CMOS, 1.3 mm ²	1.7-4.7 μ W	70-85%	Self-powered P _{IN} > 5.7 μ W	— V _O = 5 V
[65]	Piezoelectric	40 Hz (—)	405 μ W (0.31 g)	—	606 mm ³ + 54 grams	Diode bridge, Single SECE for 3 harvesters	Discrete components	8-18 μ W	50-74%	Self-powered P _{IN} > 30 μ W	— A _{IN} > 0.1 g V _O = 6.5 V
[66]	Piezoelectric	47 Hz (5 Hz)	11.9 mW (0.5 g)	—	5.3 cm ³ + 36 grams	Diode bridge, DCM flyback converter, MPPT	Discrete components	408 μ W	62-72%	Self-powered P _{IN} > 2 mW	— V _O = 5 V
[67]	Piezoelectric	145 Hz (—)	80 μ W (0.025 g)	—	—	Energy investing switched-inductor AC-DC converter	0.18 μ m CMOS, 2.3 mm ²	2-3 μ W	55-70%	Battery (2.7 V) P _{IN} > 10 μ W	~15 cm ³ A _{IN} > 10 mg V _O = 3.7 V
[68]	Piezoelectric	95 Hz (—)	250 μ W (0.5 g)	—	470 mm ³ + 0.5 grams	NPVC, DCM buck-boost	0.18 μ m CMOS, 0.05 mm ²	—	54%	Self-powered P _{IN} > 5 μ W	— V _O = 3 V
[69]	Electromagnetic	8 Hz (4 Hz)	67 μ W (1.0 g)	0.01 mW/cm ³ /g ²	4.8 cm ³	Active diodes, Dual-rail bias generator	0.35 μ m CMOS, 0.6 mm ²	0.2-5.4 μ W	81%	Self-powered P _{IN} > 5 μ W	16 cm ³ A _{IN} > 0.5 g V _O = 1.5 V
[70]	Electromagnetic	70 Hz (—)	162 μ W (0.6 g)	—	—	Active voltage doubler, MPPT, Voltage regulator	0.35 μ m CMOS, 1.3 mm ²	—	62-72%	Self-powered P _{IN} > 11 μ W	— A _{IN} > 0.16g V _O =2.2 V
[71]	Electromagnetic	66 Hz (—)	380 μ W (0.3 g)	—	—	NPVC, Active diode	0.35 μ m CMOS, 0.6 mm ²	1 μ W	80-90%	Self-powered P _{IN} > 40 μ W	— A _{IN} > 0.1 g V _O =0.5-3V
[72] [73]	Electromagnetic	10 Hz (3 Hz)	3.3 mW (1.0 g)	0.33 mW/cm ³ /g ²	9.9 cm ³	Diode bridge, AC-DC step-up with SECE	Discrete components	—	40%	Self-powered —	— V _O = 5 V
[74]	Electromagnetic	22 Hz (—)	2.5 mW (0.75 g)	0.03 mW/cm ³ /g ²	154 cm ³	Voltage multiplier with active diodes	Discrete components	23.7 μ W	65-90%	Self-powered —	— A _{IN} > 0.5 g V _O = 3.7 V

(*) SECE: synchronous electric charge extraction, NPVC: negative to positive voltage converter, MPPT: maximum power point tracking.

DCM: discontinuous charging mode, P_{IN}: minimum raw harvester output tested for self-startup and rectification, V_O: maximum rectified voltage output.

A_{IN}: minimum acceleration input tested for self-startup and rectification.

mechanical damping on the harvester displacement due to the feedback from bias-flipping [37].

A MEMS harvester with same design dimensions but with a tungsten proof mass is measured to provide 94.5 μ W (8.2 V_{PP}) raw power output on a resistive load when excited at 1 g vibration at 155 Hz (Fig. 15). Integrated with this harvester, the power management circuitry charges a 70 mF ultra-capacitor from 0 V to 1.85 V in 50 minutes (Fig. 16). In the charging period from 1.2 V to 1.85 V, the effective charging power into the ultra-capacitor is 50–55 μ W, and the charging efficiency is ~58%. After the initial charge, the ultra-capacitor voltage is kept at the same potential against its self-leakage by the continuous small current pulses from the harvester. In this case, the efficiency of the system is partly limited by the high self-leakage

of charge within the ultra-capacitor. The self-leakage currents of the 70 mF, 20 mF and 1 mF capacitors used as the permanent energy storage in this study are measured to be about 5 μ A, 1 μ A, and 0.15 μ A respectively, when tested after 10 min charging period at constant 1.85 V and 1 minute wait period at open circuit (Fig. 17).

The maximum power consumption of the chip during active operation is measured to be less than 0.5 μ W, and is observed to have a linear dependence on the supply voltage as expected due to the fixed V_{BIAS} source (Fig. 18). The power consumption in the sleep-mode is <10 pW as a result of utilizing two separate energy reservoirs in the system, so that the power management circuitry is activated only when there is sufficient vibration input. When a 20 mF or 1 mF capacitor is used as

the energy storage unit, the minimum vibration required for full system operation is measured as 0.15 g or 0.1 g, respectively, at which the MEMS harvester with tungsten proof mass can provide more than 2.7 μW raw power output. The maximum acceleration for reliable operation is tested as 1.5 g ($>200 \mu\text{W}$ raw power output), above which the high bending stress at the cantilever beam is observed to cause fatigue and a decrease of electromechanical coupling in the piezoelectric material over time.

The presented circuit architecture avoids the use of an inductive buck-boost converter to adjust the harvester output voltage, thus enables low power consumption and a minimum number of off-chip inductors. The optimal potential on the MEMS harvester with tungsten proof mass is measured to be from 1.5 V_{RMS} to 3.0 V_{RMS} at input acceleration values between 0.2 g and 1.0 g, for a resistive load. Although optimal loading of the harvester at varying operation conditions is preferable, the use of an asymptotic output voltage of 1.85 V in the power management circuit causes a calculated worst-case efficiency loss of $<11\%$ in this acceleration range, specifically for this harvester. As a future work, an improved prototype may include an optimum-load matching stage in order to dynamically define the potential on the harvester according to the acceleration, and achieve a more efficient and robust transfer of energy from mechanical to electrical domains at variable vibrations.

V. CONCLUSION

The performance of the integrated platform is compared with some of the state-of-the-art power management circuits assembled with piezoelectric or electromagnetic vibration energy harvesters (Table I). Here, the summarized studies have mostly focused on solutions for increasing the harvesting efficiency of either only the mechanical harvester [26] or only the power management circuitry [29], [32]. In this study, an efficient stand-alone energy generation platform is realized by integrating some of these efforts into a single microsystem. Here, the presented work has the benefits of hybrid integration of a high power density and voltage output MEMS harvester with an efficient power management circuit to create a compact and autonomous micro platform.

In a volume of $< 0.25 \text{ cm}^3$, the system-level integration of a piezoelectric MEMS energy harvester with its power management IC and off-chip energy storage components is achieved by chip-level vacuum packaging, use of vertical Si-vias for electrical feedthroughs, and utilization of the top glass package for electrical interconnects like a PCB board via patterned aluminum leads. The MEMS harvester relying on the thinned bulk-PZT technology can provide 94 μW raw power output under 1 g vibration at 155 Hz with a much larger bandwidth (13 Hz) than most resonant vibration harvesters.

The power management IC rectifies the sinusoidal output from the MEMS harvester with a low voltage drop of $\sim 30 \text{ mV}$ and stores the energy in an ultra-capacitor without depending on a pre-charged battery or tuned operation frequency. In a 0.25 mm^2 active footprint, the circuitry incorporates a bias-flip stage, a rectifier with an active diode, a trickle charger with temporary and permanent energy reservoirs, and a supply-independent

bias generator. The circuit consumes 10 pW and 0.5 μW in sleep and active modes respectively, and provides charging efficiency of 58–86% from the leads of the harvester to the ultra-capacitor. Overall, the entire energy harvesting platform weighs less than 0.6 grams, can harvest from excitations $>0.1 \text{ g}$ with self-startup, and is tested to charge a 70 mF ultra-capacitor from 0 V to 1.85 V in 50 minutes under 1 g input vibration.

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